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Summary

Computer Science undergraduate (AI & Robotics) with hands-on experience in AI/ML, full-stack development, data engineering/BI, embedded systems, and automation; research experience in generative AI model routing.

Technical Skills

Languages: Java, C, C++, Python, SQL, JavaScript

Web: React.js, Next.js, Node.js, FastAPI, HTML5, CSS3

AI/ML: RAG, LangGraph, Transformers, Prompt Engineering, Computer Vision, Speech AI

Data & BI: PostgreSQL, Power BI, DAX, Dimensional Modeling, ETL/ELT, MySQL, MongoDB, ChromaDB

Tools & Cloud: Git, Docker, GitHub Actions, AWS (EC2, S3, Lambda), UiPath, Arduino IDE

Core Concepts: DSA, OOP, DBMS, Operating Systems

Professional Experience

Research Intern | Centre of Advanced Data Science (CADS), VIT Chennai *May 2025 - Oct 2025*

- Co-authored **CRISP**, a prompt-routing system that scores the creativity level of incoming requests and forwards each to the most cost-effective generative model, achieving up to **35% savings** in inference spend.
- Assembled a 500-sample training set with **GPT/Gemini API** ratings and human annotation; evaluated **GAN, Transformer, and Diffusion** architectures using CLIP Score.

Application Development Intern | Integra Global Solutions *May 2025 - July 2025*

- Developed responsive UI screens for a cross-platform mobile application, translating design requirements into production-ready interfaces and improving user flow and load performance through iterative testing.

Projects

ChainSight: Supply Chain & Manufacturing BI Platform *PostgreSQL, SQL, Python, Power BI*

- Built an end-to-end PostgreSQL pipeline on a **synthetic** manufacturer-retailer dataset (250K orders, 1.2M inventory snapshots): dirty source files are cleaned in SQL staging (de-duplication, date/status repair, every rejected row logged) and modeled as a **star schema** with 13 fact tables, 15 dimensions and enforced grain.
- Wrote **81 analytical SQL queries** (CTEs, window functions, cohort retention, OEE) and **166 automated data-quality checks**; cut a query from **90 ms to 0.9 ms** via a composite index (EXPLAIN ANALYZE).
- Designed a Power BI semantic model with **200+ DAX measures** and a 10-page report specification, including a supply-chain control tower that drills from late orders back to production, materials and suppliers.

PerceiveAI: Agentic Multimodal Assistant for Visually Impaired Users *Next.js, FastAPI, LangGraph, RAG, ChromaDB, Gemini API*

- Built an agentic assistant that routes image + voice queries across **document, hazard/obstacle, scene, and everyday-task** (currency, color, label-reading) pipelines instead of a single generic vision prompt, with a swappable provider layer supporting **Gemini, Ollama, and OpenAI**.
- Achieved **100% recall and 90% precision (F1 0.947)** on a 15-scenario hazard-detection benchmark and **100% retrieval accuracy** on a document-QA benchmark, via an automated evaluator run against the live pipeline code rather than hand-reported metrics.
- Implemented source-grounded **OCR + RAG** to prevent hallucinated document answers, deterministic safety-reasoning, a continuous **Explore mode** for real-time spoken narration, and end-to-end latency instrumentation across the voice-to-response pipeline.

AutoNova Parts: Inventory Stock Alert System *UiPath, HTML/CSS/JS, jsPDF*

- Automated restocking with a UiPath bot that monitors inventory, sends supplier notifications, and triggers purchase workflows without manual intervention.
- Built browser-based demand forecasting using EMA, linear trend, and velocity scoring to predict 7-day part demand and automatically generate PDF purchase orders.

Education

Vellore Institute of Technology, Chennai *August 2023 - Expected August 2027*
Bachelor of Technology, Computer Science Engineering (AI & Robotics Specialization) **CGPA: 8.82/10**

Narayana Junior College *March 2023*
Intermediate (Class XII), MPC, TSBIE **Marks: 970/1000**